Chapter 3
Sample Collection, Preparation, and Identification of Microplastics in Soil: Issues and Recommendations

Montaha Behbehani, Saif Uddin, Yasmeen Al-Babtain, Maha Al-Sinan, and Shua'a Al-Ruwayeh

Abstract The omnipresence of microplastics (MPs) in agricultural soils, driven by practices such as compost and fertilizer addition, sewage sludge application, plastic mulching, and wastewater irrigation, is the main input of MPs to soil and poses significant risks to soil health and food safety. The paper evaluates methodologies from 39 global studies to identify gaps and inconsistencies in MP sampling, preparation, and identification protocols. Key findings reveal diverse sampling strategies, with composite sampling across depths (0–30 cm) and grid/quadrant approaches recommended to account for soil heterogeneity. Pre-treatment methods, including organic matter removal via 30% H_2O_2 or enzymatic digestion, and density separation using NaCl or NaI, vary widely, impacting MP recovery rates. Identification techniques predominantly combine microscopy with spectroscopic methods (FTIR, Raman), though challenges persist in false positives, polymer degradation, and nanoparticle detection. Methodological disparities in reporting units (e.g., MP/kg vs. mg/kg) hinder data comparability, complicating risk assessments and policy formulation. Quality assurance measures, such as procedural blanks, contamination controls, and spiked recoveries, are inconsistently applied, underscoring the need for standardised protocols. Recommendations emphasise harmonised sampling designs, optimised extraction workflows, and integration of advanced techniques like laser direct infrared spectroscopy for accurate quantification. Establishing global standards is imperative to generate reliable datasets, inform regulatory frameworks, and mitigate agricultural plastic pollution effectively.

Keywords FTIR · Raman · LDIR · Quality assurance · Risk assessments

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3.1 Introduction

The ubiquitous presence of microplastics in agricultural soils as a result of the use of sewage sludge, sludge-based fertilizers, composts, plastic mulch films, and wastewater for irrigation has emerged as a critical environmental concern for both soil health and food safety. For assessing the risks and generating intercomparable data across regions, harmonised and standardised methodologies are needed. Some efforts have been made (Withana et al. 2024; Uddin et al. 2020a) to harmonise methodologies for sampling, extraction, identification, and quantification that are essential to assess the level of microplastic contamination and ecological risks. Robust, harmonised methods for soil MP analysis are vital to enable cross-national data comparability and track progress toward environmental targets. Hartmann et al. (2019) emphasized the need to prioritise standardised protocols so countries can generate consistent and reliable data to map plastic contamination trends, inform policy decisions, and accelerate global mitigation efforts to reduce plastic pollution.

The global efforts to establish the legally binding plastic treaty underscore the need for standardised monitoring and reporting frameworks (UNEP 2025), as emphasized at the 5 United Nations Environment Assembly with the establishment of the Intergovernmental Negotiating Committee (INC). The first part (INC-5.1) of the fifth session took place from 25 November to 1 December 2024 in Busan, Republic of Korea. The second part (INC-5.2) took place from 5 to 14 August 2025 in Geneva, Switzerland.

In this chapter, we have attempted to critically evaluate the methods employed by various workers; 39 studies have been highlighted to identify the methodological differences and data quality objectives.

3.2 Sample Collection

Soil being a heterogeneous matrix requires appropriate considerations to select a representative site for microplastic determination. However, the criteria for sample site selection needs to be aligned with the research objectives (Withana et al. 2024; Uddin et al. 2020a, b, 2021, 2023; Kumar et al. 2020; Zaman and Uddin 2024).

Selecting an appropriate site for soil sampling requires a comprehensive assessment of the area's historical use, particularly practices like agricultural plastic mulching, municipal waste disposal, and irrigation with treated wastewater, as these directly influence MP presence. For a realistic assessment, both localised sources and dispersed origins should be evaluated. On large spatial scales, the topography, soil type, and land use-land cover are important considerations. The time of sample collection is also a critical factor; heavy rainfall can wash out the lighter MPs into the waterways (Schell et al. 2022). In agricultural soils where plastic mulch, sewage sludge, or compost is applied, sampling should be done before the next planting

season. If the edge effect and carryover are not the objective of the study, samples should be taken centrally within fields (Piehl et al. 2018), whereas others have suggested that samples be taken on all four corners and from the center of a quadrant to make a composite sample (Yang et al. 2021a; Beriot et al. 2021). The ISO 18400-102 is designed for soil sampling (ISO 2017) and is applicable for MPs, employing random, stratified, or targeted approaches based on research goals (Möller et al. 2020).

3.2.1 Sampling Design

Based on the review of the sampling strategies used in the studies, it is observed that most of the studies adopted a grid-based sampling strategy or a quadrant system to ensure collection of a representative sample.

Several of these studies have adopted an approach of stratified sampling by collecting samples from different depths. The most commonly sampled depths were 0–10, 10–20, and 20–30 cm (Table 3.1). Most commonly the sampled layer is shallow (0–10 cm) it is suitable for surface accumulation from mulch fragmentation, sludge, and compost applications. Some studies, like that in Schleswig-Holstein, Germany, and Xinjiang, China, have sampled 10–30 cm as they intended to look at the vertical migration of MP in the soil profile. Few have looked at different layers within the profile, like the report from Shaanxi, China, where the samples were collected at 0–10, 10–20, and 20–30 cm to assess MP distribution.

While samples are collected, it should be kept in mind that most agricultural farms undergo tilling and furrowing that often disturbs the topsoil, and the presence of MPs might not solely be due to vertical migration.

3.2.2 Soil Collection

Most of the studies utilized a quadrant-based sampling approach, usually sampling the surface layer 0–10 cm using metal or stainless steel spoons, shovels, or scoops (Withana et al. 2024; Corradini et al. 2019; Crossman et al. 2020; Ding et al. 2020; He et al. 2021; Bello et al. 2025). However, to collect a deeper soil profile (30 cm or more), sediment corers, augers, or Lenz samplers are usually used (Möller et al. 2020; Weber et al. 2021). Most researchers have excluded plastic components to prevent cross-contamination, which is essential, so samples should be stored in metal or glass containers. Depending on the tools like Pürckhauer augers may necessitate combining multiple cores from adjacent sites to make a composite sample as larger sample size is needed for representative samples for MP in soils. Alternatively, larger pile-driving probes can be used, though they will be expensive (Weber et al. 2022).

In most studies, the surface microplastic concentration is higher than the deeper layer. Liu et al. (2018) reported 78.00 ± 12.91 MP kg^{-1} in surface soil and

Table 3.1 Summary of the methodology adapted for microplastics in soil by different workers

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Nanjing and Wusi Region, China	*Soil Samples:* Soil was collected from agricultural lands, with no history of sludge application and treated wastewater use.	Surface soils were collected using a stainless steel shovel.	Samples were air-dried and sieved through 5-mm and 1-mm stainless steel sieves. Microplastics larger than 1 mm were manually removed from the sieve.	Collected from 3 WWTPs	Samples were air-dried and sieved through 5-mm and 1-mm stainless steel sieves. Microplastics larger than 1 mm were manually removed from the sieve.	30% H_2O_2 at 70 °C to remove organic matter (OM). Samples heated until nearly dry.	Solutions tested: NaCl (1.2 g/cm^3), $ZnCl_2$ (1.5 g/cm^3), NaI (1.8 g/cm^3). Shaken for 1 hr (initial) and 30 min (repeat), settled for 48 hr. Supernatant siphoned and combined.	Vacuum filtration using 20 µm pore size nylon membrane filter. PTFE and glass fiber membranes were avoided due to shedding.	MPs washed with **30% H_2O_2 + H_2SO_4 (3:1 v/v)** at 70 °C to remove residual OM.	Stereo microscope: Observed shape (fiber, fragment, bulk), color, and size (0.02–5 mm). ImageJ is used for size classification. *µ-FTIR (Micro-Fourier transform infrared spectroscopy):* ATR mode, 4 cm^{-1} resolution, 32 scans. Spectra were compared with those in the Hummel Polymer and Additives databases. Three random 1x1 cm^2 areas were analyzed per membrane.	MP abundance: items/kg (soil or sludge).	*Contamination prevention*: Glassware only, rinsed with ultrapure water. Researchers wore cotton lab coats. Flasks covered with aluminum foil during flotation. *Membrane selection:* Tested quartz, glass fiber, PTFE, and nylon membranes; selected nylon to avoid fiber shedding. *Replicates:* Three parallels per treatment. *Oxidation validation:* Tested virgin polymers (PE, PP, PS, PA, ABS, PET) to ensure no degradation during oxidation. *FTIR verification:* Non-plastic particles (e.g., black carbon) were excluded via spectral analysis.	Li et al. (2019)

Shanghai, China	20 vegetable farmlands were samples GPS coordinates recorded for each site.	0–3 cm shallow layer and 3–6 cm deeper layers, measured with calipers.	1 kg of soil per sample, visible debris >5 cm removed Stored in aluminum boxes to avoid plastic contamination.			Dried soil at 70 °C for 24 hours.	Density separation using NaCl (1.19 g/cm^3) and ultrasonicated for 2 minutes. Stirred for 30 minutes, settled for 24 hours.	Filtered through 20 µm nylon membrane filter. Repeated 3 times for complete extraction.	Treated supernatant with 30% H_2O_2 at 50 °C for 72 hours	Visual classification using stereomicroscope (Nikon SMZ25) for shape (fiber, fragment, film, pellet) and color. Size categories: microplastics (<5 mm), mesoplastics (5 mm–2 cm). Polymers identified using µ-FTIR In transmittance mode (4000–675 cm^{-1}, 32 scans). Spectra were matched to Hummel Polymer database (≥70% match threshold). Ninety-nine randomly selected particles were analyzed.	Microplastic/ mesoplastic abundance in items kg^{-1} of dry soil.	ForrRecovery test samples were spiked with PP, PE, PA, PET, PVC, PC, ABS, PMMA, and PS microplastics. Recovery rates were > 90% for PP, PE, PA, PC, ABS, PMMA, and PS; PET and PVC not recovered due to its density higher than NaCl. Glass/metal equipment rinsed with Milli-Q water. No plastic containers used. Researchers wore cotton lab coats and gloves. Samples stored in closed containers to avoid airborne contamination. Triplicate blank tests confirmed no microplastics in controls.	Liu et al. (2018)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Shangsuan, Anle, Dunshang, Dagoujian, and a buffer zone, China	Five agricultural soils in the Dian Lake basin	Collected six sub-samples per plot using a narrow spade from two depths 0–5 cm and 5–10 cm	Composite samples were sieved through a 10 mm mesh. Air-dried at 25 °C, roots and large debris removed.	Composite agricultural soil	Wet-sieving of air-dried soil through 2.00, 0.25, and 0.05 mm sieves. After that, the samples were centrifuged for 10 minutes at 2300 rpm to separate aggregates. The >2 mm were considered as large macro-aggregates, 0.25–2 mm as small macro-aggregates, 0.05–0.25 mm as micro-aggregates.	Samples treated with 35% H_2O_2 and 10% $FeSO_4$ at 50 °C to remove organic matter	Saturated NaI with a density of 1.8 g/cm^3 solution for density separation.	Supernatant filtered through stainless steel meshes (1, 0.25, 0.05 mm).		Visual classification using dissection microscope at 40× Categorized into size range of 10–1 mm, 1–0.25 mm, 0.25–0.05 mm and shapes were identified as fiber, fragment, and film, while pellets and foams were excluded	No polymeric characterisation as done.		Zhang and Liu (2018)

Xinjiang, China. Quadrat Sampling	Composite samples from a quadrat measuring 200 cm × 100 cm × 30 cm depth from 593 sites were collected.	Soil samples were manually collected from the top 30 cm.	A questionnaire was sent to 672 farmers, 593 responded giving information of crop type, mulch film thickness, mulching years, etc.	Ultrasonic cleaning to remove soil particles.	Blotting paper is used to absorb moisture. The samples were then air-dried and weighed.					Visual identification was done and no spectroscopic methods were used.	The results were reported as film abundance in kg ha^{-1} for individual sites and in Metric tons for the regional total	Large sample size to reduce bias. Ultrasonic cleaning to ensure accurate weight measurements. Compared results with national standards (GB/T 25413-2010). But no mention of procedural blanks or contamination controls during lab analysis. And potential self-reporting bias in questionnaire data.	Zhang et al. (2016)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Szeged, Hungary	Soil samples from greenhouse and farmlands. The area was divided into 15 areas of 52 × 9 m and Three areas were randomly selected for sampling.	Soil samples were collected from two depths, i.e., 0–20 cm and 20–40 cm using metallic auger	Soil samples were oven-dried, homogenized, sieved and composite samples were used. Visible macroplastic debris on the surface was hand-picked by two observers. The collected plastics were precleaned and stored in large polyethene bags.			Organic matter was digested using 30% H_2O_2 and Fenton reagent	$ZnCl_2$ was used for density separation.	Extracts were filtered through nylon and Whatman filters, dried. Macroplastic materials were soaked in tap water for 48 hours, washed, sieved, dried, separated, counted, and measured.		Microscopic examination of filters. Suspected microplastic particles were confirmed using the hot needle and Raman spectroscopic analysis.	Macroplastics were reported in kg ha^{-1} and categorized by size (0.5–1.0 cm, 1–5 cm, 5–10 cm, 10–15 cm, >15 cm). Microplastics were reported as MP kg^{-1}	Cleaning the auger before each sampling, avoiding sample mix, and using minimal plastic materials. Rinsing laboratory apparatus with distilled water three times, wearing cotton lab coats and gloves, and covering samples with aluminum foil to prevent contamination.	Sa'adu and Farsang (2022)

Shenyang Agricultural University, China	Arable land	Soil cores were taken from random locations in each plot using a coring drill of 8.5 cm diameter from 0–10 cm, 10–20 cm, 20–40 cm, 40–60 cm, 60–80 cm, and 80–100 cm.	Soil samples were sieved through a 5-mm metal mesh to remove macroplastic residues and roots, and air-dried.			20 g of air dried soil samples was mixed with density separation solution, sonicated, filtered, and treated with 30% H_2O_2 solution, heated, and filtered again..	$ZnCl_2$ solution was used for density separation	The supernatant was filtered through a membrane filter, dried		Stereo microscope used for identification. The polymeric characterisation was done using μ-FTIR. Microplastics were classified into films, fibers, and granules.Size, and color were also reported	MP $m^{-2}$for macroplastics and MP kg^{-1} of soil for microplastics	Soil stored in cotton bags. Metal and glass was used and washed with pure water, and covered with aluminum foil. All the solutions were filtered, and cotton lab coats were worn. Three replicates without soil were run concurrently to check for potential plastic contamination during extraction. No plastics were detected in these blank control samples. Positive control samples were used to test the efficiency of the extraction method, achieving recovery rates of 87% ± 3% for film microplastic and 90% ± 7% for fiber microplastic.	Li et al. (2022)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
China	Experimental soil where PS was added		Seeds were placed in sterilized beakers with PS-MP solutions and grown in a climate-controlled atmosphere for 14 days at 25 ± 1 °C and 75% humidity.							Transmission electron microscopy (TEM) and scanning electron microscopy (SEM) were used to observe PS-MP distribution in rice seeds, roots, and buds.	mg L^{-1}	Samples were fixed, rinsed, dried, and coated for SEM observation. TEM samples were embedded, sliced, and dispersed on copper mesh.	Zhang et al. (2021)
Changsha, China	Agricultural soil	0–20 cm	The soil was air-dried, debris removed and passed through a 10 mm steel sieve, and stored in dark conditions										Yang et al. (2021a)

Ishikawa Prefecture, Japan	19 paddy fields in the Tedori River alluvial fan, Five points per field (four corners and centre) using a 15-cm-diameter core sampler	0–15 cm, divided into two layers (0–10 cm and 10–15 cm). 1 kg soil sample was also taken from the centre for physico-chemical analysis.	The soil samples were dried at 60 °C for 2 days			Wet Sieving was performed by mixing with tap water, stirring, settling overnight, sieved using 1-mm mesh.		Sieved through 1 mm sieve	Visual Sorting of Microcapsules based on the criteria: 2–3 mm. and color White/ yellow discoloration, spherical hollow capsules.	SEM used for analyses of surface morphology of microcapsules. FT-IR for polymer identification	Reported in mg/ kg soil and g/m^2. The results were also reported as pieces/kg soil and pieces/m^2.	Two additional individuals verified visual sorting to minimize errors. Spiked samples were used and 85–95 recovery	Katsumi et al. (2021)
Tainan City, Taiwan	Five farmlands (A–E) with different crops and mulch. Each farm is divided into three subplots; composite samples collected from, five subsamples per subplot	0–5 cm top layer and 20–25 cm bottom layer	Air-dried, homogenized, and sieved using 1 and 5 mm stainless-steel sieves			30% H_2O_2 used overnight to degrade organic material	NaCl solution, stirred, settled for 2 hours, and	Filtered through 1 µm glass microfiber filters.	Stereomicroscope used to classify MPs by shape (fibers, fragments, films, foams, pellets, microbeads) and color.	ATR-FTIR used for polymeric identification	MP/m^2	Control farm as a background control. Control Soil sample ignited at 500 °C to remove impurities tested for contamination during transportation. Spiked samples were also analyzed	Fakour et al. (2021)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Sulur, Nagondapalli, Royakottah, Krishnagiri regions, Tamil Nadu, India	Agricultural soils from areas that have been mulched for 2–10 years	0–10 cm, 11–20 cm, and 21–30 cm.	A stainless steel probe was used to collect 15 soil cores, composited. Control samples (virgin soil) taken 500 m away		The equipment was cleaned with 70% ethanol. Samples were air-dried, powdered, and stored at 4 °C. Soil sieved using stainless steel sieves (pore sizes: <5 mm to <1 mm).	Digested with 30% H_2O_2 at 50 °C for 72 hours.	Floating organic matter separated via saturated NaCl solution.	Filtered, dried, and plastic weight calculated as: $W = w_1 - w_2$) (initial and final filter weights)		Stereo microscopy, shapes identified as films, fibers. Further identification of surface morphology and elemental composition was done using SEM-EDX	g kg^{-1} soil and also MP/kg soil	Glass containers, nitrile gloves, and cotton lab coats were used. Virgin soil samples for baseline comparison. Five replicates for enzyme activity assays. The equipment was sterilized with 70% ethanol between samples.	Kumar and Sheela (2021)
Yeoju City, South Korea	100 sites covering forest (n = 8), traffic/ roadside (n = 19), residential (n = 5), and agricultural (n = 68) areas. Subdivided into orchard, paddy field, upland, and greenhouse. Systematic 2.5 km × 2.5 km grid.	0–5 cm.	Hand auger used to collect three replicates per site. Samples stored in stainless steel containers. Oven-dried at 60 °C for 48 hours. Mesoplastics separated using a 5-mm sieve.			Treated with $FeSO_4$, H_2SO_4, and H_2O_2, heated at 75 °C for 24 hours	Microplastics (<5 mm) extracted using $ZnCl_2$ solution (1.7 g cm^{-3}), stirred, settled, and repeated thrice.	Filtered through 0.45 µm cellulose acetate membranes		Digital stereo microscope used to count and classify by shape (fragment, film, fiber, sphere) and color (black, red, green, etc.). Polymers identified using FTIR by matching spectra to reference databases.	MP kg^{-1} soil, categorized by size: <1 mm, 1–5 mm, >5 mm and Polymer Types reported as percentages (e.g., SBR: 43%, PP: 22%).	Stainless steel tools and containers used. Triplicate samples per site. FT-IR analysis on 60 representative samples to confirm polymer composition.	Choi et al. (2021)

Swiss flood plain, Switzerland	Remote high mountain soil and from populated regions	Collected from topsoil layers (depth unspecified) using standardized soil sampling protocols. Mixed samples from each site were analyzed (three replicates per site).			Custom-built setup involving a vacuum pump, magnetic mixer, and aeration system	Tested HNO_3 (90 °C, 48 h) for organic matter digestion	Combined with centrifugation; rubber discs are used to prevent resuspension of mineral particles.	Particles filtered through anodisc membranes (0.45 µm porosity).		Polymer identification using Raman spectroscopy (e.g., PP, PE, PVC, PET). FT-IR validation.	MP/kg^{-1} of soil, and categorized by size as small microplastics (<1 mm), Large microplastics (1–5 mm), Mesoplastics (>5 mm)	Anodisc filter to rule out contamination. Recovery tests were conducted. Triplicate samples per site. Chemical resistance testing after HNO_3 treatment. Negative controls.	Scheurer and Bigalke (2018)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Swiss Seeland, Switzerland	11 agricultural drainage sites (8 passive flow systems, 3 basin systems)	Flowing water was directly collected into 3 L pre-cleaned glass containers. Basin water was sampled with a pre-cleaned metal bucket, transferred to glass containers. 5 sites sampled to assess temperal variability.				Backwashed with 15 mL 60% H_2SO_4 for 30 minutes. Filtered through 8 μm cellulose nitrate membrane filters	Centrifugation (3000 RPM, 30 min) to isolate MP	Supernatant filtered through 0.2 μm Anodisc filters for FTIR analysis		FTIR spectroscopy, particles >100 μm analyzed.	MP L^{-1} with mean ± standard deviation.	Processed 2.5 L Milli-Q® water; as a blank with mean blank = 0.8 MP L^{-1}. Detection limit: Mean blank +3SD = 2.2 particles L^{-1}. Recovery test with 100% recovery of spiked PE, PP, and PVC particles (1–2 mm). One sample in triplicate, one in duplicate; mean RSD = 54%. Temporal replicates (same site, different dates): mean RSD = 72%. For minimizing contamination lab equipment rinsed with Milli-Q® and ethanol; sample preparation under HEPA-filtered laminar flow hood.	Bigalke et al. (2022)

Speyer, Germany	Experimental agricultural farms where 190 T of sludge per hectare was applied										MP g^{-1}		Tagg et al. (2022)
Ilia County, Greece	Agricultural fields and Five uncultivated "blank" samples	0–30 cm using stainless-steel shovel	Air-dried, sieved through 5 mm mesh, stored in aluminum containers			Treated with 30% H_2O_2 at 70 °C for 2 days to remove organic matter.	Saturated NaCl (1.18 g/cm^3) mixed with 150 g soil, stirred, rested, and filtered through 500 µm and 20 µm sieves. Repeated until no floating material			Stereo microscope to identify 20 µm–5 mm. ATR-FTIR to identify polymer type. SEM for analyzing surface degradation of MPs.	MP kg^{-1} classified into 2 classes 20 to 500 µm 500 µm to <5 mm	Blank samples. Triplicates for each sample; Stainless-steel tools, aluminum containers, visual exclusion of non-BMF MPs. To avoid contamination.	Isari et al. (2021)
Murcia, Spain	3 fields sampled that had a Long-term use of LDPE, PAC, and biodegradable mulch over 12 years	0–10 cm and 10–30 cm depths			10 g soil or 5 g compost mixed with water, centrifuged (3000 rpm, 10 min), filtered through 2.5 µm filter.	Treated with H_2O_2 and heated (130 °C) to differentiate MPs from organics.				Visual examination using a Stereo microscope. Particles are classified by shape, color, and heat response. ImageJ software is used to quantify particle count and size (30 µm–2 mm). FTIR spectroscopy to identify polymers	MP kg^{-1} and mm^2 kg^{-1} (particle area).	Filed blank samples; Three replicates per sample. Instrument Validation. Detects only light-density MPs >30 µm. Potential underestimation of non-floating plastics	van Schothorst et al. (2021)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Netherlands	2 farms with loamy sand, 7 years of compost; clay, 20 years of compost). Each farm had 3 fields, 5 points per field	0–10 cm and 10–30 cm depths		Municipal organic waste and garden/ greenhouse waste	10 g soil or 5 g compost mixed with water, centrifuged (3000 rpm, 10 min), filtered through 2.5 µm filter.	Treated with $H_2O_2$and heated (130 °C) to differentiate MPs from organics.				Visual examination using a Stereo microscope. Particles are classified by shape, color, and heat response. ImageJ software is used to quantify particle count and size (30 µm–2 mm). FTIR spectroscopy to identify polymers	MP kg^{-1} and mm^2 kg^{-1} (particle area).	Filed blank samples; Three replicates per sample. Instrument Validation. Detects only light-density MPs >30 µm. Potential underestimation of non-floating plastics	van Schothorst et al. (2021)
Winterthur, Switzerland	Processed organic waste from municipalities, gardens, and households	Samples at three stages: incoming organic waste (conveyor belt), sieving residues, and final solid digestate.	Sampling was conducted over five days (June–August 2017), 15 samples total (4.5 L each, collected every 40 minutes over 2 hours).		Bulk density measured for compost and digestate samples.	Washed samples under running water over a 6-mm mesh to separate plastics. Dried retained material at 40 °C for 4 days.				Visually inspected dried samples to collect plastic pieces >1 mm. Microscopic examination and classification into plastic bags, packaging materials, etc. FTIR spectroscopy for identification of polymers.	Tonnes yr^{-1} for material flow analysis. Plastic Concentrations as a percentage of wet organic waste in % and total accumulation in Swiss soils reported in tonnes	Monte Carlo iterations to quantify variability and uncertainty in plastic content	Kawecki et al. (2021)
Arusha, Tanzania	Soils from irrigated farms									ATR-FTIR			Kundu et al. (2022)

Mauritius	Agricultural soils collected from three sites	0–10 cm and 10–20 cm	Each site is divided into a 20 × 20 m quardant. 40 samples per grid composited into a single sample. Samples were dried, sieved, and analyzed	Grab sampling from three WWTPs. Sludge (1 L) was collected after primary/ secondary treatment; wastewater (500 mL) was collected before discharge. Samples were stored in glass bottles.	Sludge samples were mixed with water, NaCl, centrifuged, and digested with H_2O_2. The effluents were directly mixed with H_2O_2, digested, and filtered.	H_2O_2 digestion (7 days) to remove organic matter.	Flotation with distilled water and NaCl (1.19 g/cm^3), followed by centrifugation	Filtered through 0.45 µm cellulose nitrate membranes.		Microscopic examination to categorize MPs by shape (fiber, fragments, flakes, films, spheres), color, and size (0.25–5 mm). ATR-FTIR for polymer identification of particles >0.5 mm.	MP kg^{-1} for soil. MP L^{-1} for effluent MP kg^{-1} wet weight for sludge	Spiked samples with known plastics to calculate recovery rates; three replicates. Blank samples analyzed. Lab equipment rinsed with ethanol; use of gloves and aluminum foil used to cover glassware.	Ragoobur et al. (2021)
Tunisia	4 Agriculture soils with organic farming, control; Mulched; Greenhouseand soil irrigated with treated wastewater.	10–20 cm depth	Earthworms (*Eisenia andrei*) were exposed to soil spiked with MPs (<100 µm) for 7 and 14 days.			MPs in earthworms were digested using 10% KOH.	Density separation using saturated NaCl (1.19 g.cm^{-3}). The earthworm samples were density-separated by KI	Sieved through 300 µm mesh to retain particles (1 mm–300 µm). Earthworm samples were filtered through 0.22–3 µm cellulose nitrate filters.		Microscopic examinatio for color/shape. FTIR and Raman microspectroscopy for polymer identification	MP kg^{-1} in soil. MP g^{-1} tissue for earthworm	Rinsing equipment with Mill-Q water. Blank samples. 10 replicates	Boughattas et al. (2021)

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Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Chile	Agricultural soil with varying degrees of sludge application histories (0–5 applications over 10 years) from 31 fields and a Control site (no sludge) applied	0–25 cm sampled with a metallic auger; three replicates per field	Dried at 40 °C, sieved with <2 mm sieve, and stored in PET jars	Seven samples collected from fields undergoing sludge applications during the study	Dried, milled, sieved (<1 mm), and stored I PET jars		Sequential use of NaCl (5 M, $\rho = 1.20$ g cm^{-3}) and $ZnCl_2$ (5 M, $\rho = 1.55$ g cm^{-3}) solutions to float microplastics	5 g of soil and 1 g of sludge is mixed with deionized water, stirred, centrifuged, and filtered.	Supernatants filtered through Whatman No. 42 filters (retention >8 μm).	Microscopic visual identification and classified into fibers, fragments, films, or pellets based on morphology. No spectroscopic confirmation was done.	MP g^{-1} in soil/ Sludge (DW). MP weight estimated using particle dimensions and assumed polymer densities.	Reagent blanks were included in each batch to detect lab contamination. Ignited soil samples tested for transport/ storage contamination and recovery test. Five samples were analyzed five times; the coefficient of variation = 12%.	Corradini et al. (2019)

Ontario, Canada	Soil from four agricultural fields (three biosolid-treated, one control)	0–5 cm, 5–10 cm, and 10–15 cm using a stainless steel corer (5 cm diameter)	Soil cores wrapped in aluminum foil, frozen, then oven-dried (≤65 °C); biosolids refrigerated before drying	Biosolidswere collected directly from application trucks/hoses during spreading		Fenton's reagent used to remove organic content.	Sequential use of sodium tungstate (Na_2WO_4) solutions to float microplastics (MPs)	Isolated MPs filtered onto glass fiber filters		Microscopic examination to classify MPs by shape. Polymer identification using ATR-FTIR for particles >300 µm and µ-FTIR for smaller particles/ fibers.	MP kg^{-1} for soil and Biosolids reported in MP kg^{-1} and MP L^{-1}. Mass estimationusing particle dimensions (e.g., fragment volume as ellipsoids, fiber volume as cylinders.	Aluminum foil covers during drying. Spiked samples (PE, PVC, PET fibers) showed recovery rates of 80–97%. Lab coats and controlled airflow in processing areas. Procedural blanks were analyzed to account for background contamination. Triplicate analysis of composite soil samples; standard errors reported.	Crossman et al. (2020)
Mexico	Arable land												Hernández-Gutiérrez et al. (2021)
Shaanxi, China	Agricultural soils	0–10 cm	6 composite samples per site, each composed of 3 sub-samples collected 3 m apart to a 2 kg quantity. Stored in clean metal bottles and refrigerated		The soil was oven-dried at 70 °C for 24 hours.	H_2O_2 was used for organic matter digestion. Heated at 65 °C, 12 hours.	50 g soil mixed with NaCl (1.2 g/cm^3) and $CaCl_2$ (1.5 g/cm^3) solutions. The supernatant was collected after 24 hours.	Filtered through 0.45 µm filter paper, dried, and stored for analysis.		Microscopic examination to identify type, size, and shape. Followed by FTIR spectroscopy for polymer identification SEM for surface characterization.	MP kg^{-1}	Glass/metal tools used. Equipments rinsed with ultrapure water and samples wrapped to avoid airborne fibers. Blank experiments confirmed no MPs in controls.	Ding et al. (2020)

(continued)

Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Tibetan Plateau, China	17 farms	(0–3 cm and 3–6 cm layers)	Visible debris (>3 cm) was removed; samples were stored in aluminum foil bags		Sediment/soil were oven-dried (70 °C, 24 h), sieved (2 mm)	Digested with 30% H_2O_2 and 0.05 M Fe(II) solution at 75 °C for 72 hours to degrade organic matter.	Density separation using saturated NaCl (1.2 g/cm^3). Supernatants collected three times for thorough extraction.	Filtered through 0.45 µm membranes		Microscopic examination to classify MPs by shape, size, and color. Raman spectroscopy for polymer composition	MP Kg^{-1}	Tools rinsed with ultrapure water; glassware heated at 400 °C; laminar flow cabinets used during experiments. Triplicate Milli-Q water blanks analyzed.	Feng et al. (2020)
Schleswig-Holstein, Northern Germany	Farmland 15 sites	0–10 cm, 10–20 cm, and 20–30 cm.	Boreholes were drilled at four corner points of a 40 × 40 m square per site. 540 soil samples (36 per site) totaling ~123 kg of soil. Stored in glass containers to avoid contamination		Soil soaked in water, homogenized and sieved through a 1 mm steel sieve to retain particles >1 mm.	For organic matter removal the aggregates dissolved manually; excess water removed with syringes				Sieved material was examined under a stereomicroscope for MP (1–5 mm) and classified by shape and color. FTIR spectroscopy for polymer composition.	MP kg^{-1}	Glass/metal tools used; lab coats worn; instruments rinsed between samples. Spiked soil samples used for recovery assessment and blank samples analyzed.	Harms et al. (2021)

Xinjiang Uygur Region, China	Agriculture fields with cotton plantation and 5, 15, and 24 years of mulching.	0–5, 5–20, 20–40 cm	Five 1 m × 1 m plots per field, triplicate soil cores at different depths and stored in aluminum bags			Digested with H_2O_2	Density separation using NaI (1.85 g/cm^3). Air-dried soil mixed with NaI, agitated, settled, and supernatants collected. Flotation was repeated thrice.	Filtered through 7 µm glass fiber filters,					Huang et al. (2020)
Schleswig-Holstein, Northern Germany	Farmland boreholes were drilled at four corners of a 40 × 40 m agriculture fields.	0–10, 10–20, and 20–30 cm.	Soil soaked in water homogenized and sieved through a 1 mm steel sieve to retain particles >1 mm.			Organic matter removal was done by manually dissolving the aggregates and excess water removed with syringes.				Sieved material was examined under a stereomicroscope for MP (1–5 mm) and classified by shape and color. The polymers were confirmed using an FTIR spectrometer.	MP kg^{-1}	Stored in glass containers to avoid contamination. Glass/metal tools used; lab coats worn; instruments rinsed between samples. Spiked samples for recovery calculations. Blank samples were also analyzed.	Piehl et al. (2018)
Shouguang, Shandong and Xuzhou, Jiangsu, China.	Agricultural soils	0–10 cm	Plant debris was removed, air-dried, ground, and sieved with a 1 mm sieve.			Digestion with 30% H_2O_2 at 70 °C to remove organic matter.	NaCl (1.2 g/cm^3) was used for density separation.	Filtration through 20 µm nylon membranes		Identification of polymers using laser direct infrared (LDIR) spectroscopy in the size range of 20–1000 µm.	MP kg^{-1}	Avoided plastic tools; cotton lab coats worn; glassware pre-washed with HNO_3, acetone, or hexane. Blanks processed. Triplicate extraction.	Li et al. (2021)

(continued)

Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Yingstan, Jiangxi Province Southeast China	Agricultural plots with long-term repeated pig manure application	0–20 and 20–40 cm	Plant debris removed; samples air-dried, homogenized, and sieved (e.g., ≤5 mm or ≤1 mm)			30% H_2O_2 to degrade organic material.	NaCl saturated solution for density separation	Filtration through a 0.45 μm glass fiber filter		A stereo microscope was used to count and categorise particle morphology and polymers were identified by using FTIR.	MP kg^{-1}		Yang et al. (2021b)
Murcia, Spain	6 vegetable fields	0–10 cm	Each field had 3 parcels, with 3 sampling points per parcel.	Sheep faces was collected	Samples were air-dried.	High organic matter in faeces hindered complete particle recovery.	Density separation using distilled water	Soil was shaken, centrifuged, and supernatant filtered. Ultrasonic bath used to break aggregates.		Microscopic examination and images magnification). Images were taken before/after heating (130 °C for 10 s) to identify plastics via shape, color, brightness, and heat response.	MP Kg^{-1}	Using stainless steel tools. Blank samples reported no contamination. However, no recovery tests performed.	Beriot et al. (2021)
Youg-In Province, Korea	Agricultural soil										MP Kg^{-1}		Kim et al. (2021)

Paotai Town, Shihezi City, Xinjiang, China.	Agricultural soil with varying period of mulching 0, 5, 15, and 30 years.	0–20 cm	Five samples per site were collected using a multipoint mixed method from a 0.1 × 0.1 m. The samples were air-dried, sieved through meshes (2 mm, 0.9 mm, 0.28 mm, 0.15 mm), and homogenized			98% H_2SO_4 treatment to oxidize organic debris.	Water was used to disintegrate soil aggregates and density separated by distilled water.	Vacuum filtration with quantitative filter paper.	Electrostatic separation by rubbing glass rods against silk to collect small particles.	SEM was used to identify MPs and ATR-FTIR used for polymeric identification	Mg kg^{-1}	Recovery rates tested with spiked samples. Stainless steel/ glass tools, lab coats, gloves, deionized water rinsing. Unmulched soil samples processed as control and Triplicate experiments for extraction efficiency.	Li et al. (2020)

(continued)

Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Gansu and Xinjiang, Northwest China:	Farmland with varying degrees of mulching	0–10, 10–20, and 20–30 cm.	Collected manually from three quadrats (100 cm × 50 cm × 30 cm) per field, divided - Microplastics (MiPs):** Soil samples (1 kg per layer) collected from pit sides, homogenized, air-dried, and sieved (2 mm).				Extracted via flotation in distilled water, centrifugation (14,400 × g, 5 cycles)	filtration (<3 μm filter), and thermal treatment (130 °C heating to melt LDPE particles)		Microscopic identification before and after melting; smallest detected size: 0.49 mm.	MP kg^{-1}	Stainless steel/ glass tools, PP bags for storage, cotton lab coats, and gloves. Unmulched field samples were processed as control, additionally distilled water samples were processed as blank during MP extraction. Triplicate samples analyzed. Method restricted to low-density plastics and did not detect high-density plastics	Meng et al. (2020)

Shouguang City, China.	Agricultural soils	0–5, 5–10, and 10–25 cm	Samples were collected using a stainless-steel shovel		Samples were sieved (5 mm) to remove stones and debris.	Fenton reagent was used for organic matter digestion, under pH 3–5 and <40 °C (ice bath)	200 g soil was mixed with saturated NaCl solution (1.2 g/cm^3), ultrasonicated, and left to settle. Supernatants were collected after 24 hours.	The solutions were centrifuged, and MPs were filtered onto 20-µm glass fiber filters.		A stereo microscope was used for size, shape, and color identification. µ-FT-IR was used for polymer identification.	MP Kg^{-1}	Non-plastic tools, aluminum foil covers, cotton lab coats. Quartz sand was processed as laboratory blanks. Spiked samples were used for soil recovery rates. Triplicate samples were used for recovery estimations.	Yu et al. (2021)
Hangzhou Bay, China.	Farm soil Collected from 15 areas	0–10 cm	Samples were collected using a stainless shovel.			Organic matter was digested with H_2O_2 (30%, at 70 °C)	The density separation was done using NaI, ρ = 1.6 g/cm^3)	Dispersed soil in sodium hexameta-phosphate, followed by saturated NaCl solution with continuous airflow to float microplastics and filtered onto 20 µm glass fiber filter.		Stereomicroscope used to identify MPs and Polymer identification using µ-FT-IR.	MP Kg^{-1}	Non-textile lab attire, glassware, and dedicated microplastic lab. Deionized water was analyzed as blank. Spiked samples were used to calculate recoveries. Duplicates were analyzed.	Zhou et al. (2020)

(continued)

Table 3.1 (continued)

Country/ region	Sampling			Sludge Samples		Sample extraction				Identification	Units reported	Quality control	
	Matrix	Sampling depth	Pre-treatment	Matrix	Treatment	Pre-digestion	Flotation	Filtration	Post-oxidation				Reference
Valencia, Spain (11 with sewage sludge applications, 5 without) sampled.	Agricultural soil and sewage sludge	0–10 and 10–30 cm		Collected from 4 wastewater treatment plants	Samples were dried at 40 °C for 72 hours		NaI was used for density solution of heavier density MPs and lighter MPs were seperated using centrifuge with distilled water at 3000 RPM for 10 minutes.	Filtered using 11 µm pore size.		Microscopic examination done pre-/post-heating (120–130 °C). MPs were identified by increased circularity, transparency, and shininess. µFTIR analysis for polymer identification	MP Kg^{-1}	Cotton lab coats, dedicated microplastic lab. Deionized water was analyzed as blank to correct the results. Spiked samples were also analyzed to assess recovery. Duplicates were also analyzed. Organic matter interference noted; no natural blank samples available.	van den Berg et al. (2020)
Guangxi Province, China	Agricultural soils from citrus orchards	0–5, 5–15, and 15–25 cm.		Dewatered sludge from 3 wastewater treatment plants		Digestion with 30% H_2O_2 at 65 °C for 72 hours.	Density separation using $ZnCl_2$ and NaCl solutions.- Ultrasonication (15 min) to remove adhered particles.	Filtration through 25 µm and 0.45 µm filters.		Stereomicroscopic identification based on morphology and µ-FTIR for identification of polymer	MP Kg^{-1}	Cotton lab coats, and nitrile gloves were used. Containers rinsed with ultrapure water; openings covered with aluminum foil. Blank samples were analyzed as well as triplicates.	Zhang et al. (2020)

62.50 ± 12.97 items kg^{-1} in the deeper layer. However, it should be considered that agricultural soils are often tilled, so the top layer is most likely not undisturbed (van den Berg et al. 2020). On the other hand, nonagricultural, coastal, or floodplain soils are usually undisturbed, and the top 5 cm is representative of surface MP contamination in soil (Piehl et al. 2018; Xu et al. 2020).

3.2.3 Soil Sample Handling

A large sample volume of 0.5–2.0 kg should be collected to enhance representativeness and accommodate replicates for moisture, recovery rates, and physicochemical testing (Möller et al. 2020). Post-collection, samples should be stored in non-plastic containers, e.g., glass jars, stainless steel jars, aluminum foil trays, and sealed to prevent contamination (Zhou et al. 2020). Containers must be clean, dry, and labeled with location, depth, date, and collector details, and preferably stored at 4 °C.

3.2.4 Pre-Treatment

The soil samples are to be cleaned by removing roots and large debris, prior to drying and processed to generate standardized moisture-free baseline data for MP in soil. Different research groups have used varied strategies from air-drying at room temperature to oven-drying and freeze-drying. Details are summarized in Table 3.1. Liu et al. (2018) dried the samples at 70 °C for 24 hours, while van den Berg et al. (2020) followed ISO standards and samples were dried at 40 °C for 72 hours. Several studies have highlighted the deterioration of MP due to higher temperatures (Uddin et al. 2020a, b, 2022a, b, 2024; Jinadasa et al. 2023; Lee et al. 2024). Deterioration is reported in polyethylene (PE) and polyamide (PA) just over 50 °C, while 70 °C is generally safe for most other polymers (Hurley et al. 2018). Hence, to minimize structural changes, drying temperatures should remain below 60 °C, preferably ≤40 °C. Freeze-drying, though an alternative, is less practical due to limited throughput (Dorau et al. 2023) and often the clay-rich soils become too hard (stone-like) making the MP extraction cumbersome and impractical.

Sieving is performed by most workers; however, the choice of sieves is often varied. Some researchers have used a cascade of sieves to fractionate different size fractions, while others have used a 2 mm sieve that allows smaller particles to pass and retains those >2 mm. The MPs in the 2–5 mm range can be visually isolated. A major challenge lies in retaining nanoparticles and fine MPs during separation, as their small size increases the risks of loss or misclassification, thereby skewing quantification (He et al. 2018). The different approaches used for sieving have been summarized in Table 3.1. The study from China (Nanjing) and Switzerland used coarse sieving with 1–5 mm sieves to remove debris and MPs. In another study from

Dian Lake Basin, China, a fine sieve of <1 mm was used, retaining MPs and excluding aggregates. In studies reported from Spain and Taiwan, the debris, including roots, stones, and macroplastics, was manually removed. Enhancing recovery requires refined techniques, such as employing finer meshes or specialized filtration membranes optimized for capturing sub-millimeter MPs. These adjustments improve accuracy in detecting and characterizing MPs, offering deeper insights into their prevalence and behavior in soil ecosystems.

3.3 Sample Preparation

The agricultural soil, especially where the sewage sludge, compost, and fertilizers are applied, is organic-rich, and for accurate quantification of MP in soils, organic matter (OM) removal is a crucial step (Uddin et al. 2020a, 2021, 2022a, b, 2024; Lee et al. 2024; Naji et al. 2021). The organic matter removal should precede the density separation. It has been observed that sonication and density separation often are inefficient as MP embedded in organic matrixes or OM adhered to MPs like PET, nylon, etc., have a similar density range (1.0–1.4 g cm^{-3}) (Bläsing and Amelung 2018). It is quite complex to remove OM without compromising the integrity of polymers, as some of the processes are exothermic, and poorly controlled processes (when temperature gets very high) can compromise the integrity of plastic polymers. In the case of agricultural soils, the OM is also enriched with diverse plant and animal residues that have very different properties and react differently. The more reactive fraction of OM includes rapidly decomposable components such as lipids, proteins, and sugars (Bronick and Lal 2005), which can interfere with MP detection and must be eliminated during sample pretreatment. Organic matter removal typically involves chemical treatment with acids, alkalis, oxidizers, or enzymatic digestion (Bläsing and Amelung 2018; Fischer and Scholz-Böttcher 2017; Loder et al. 2017; Steinmetz et al. 2020; Thomas et al. 2020).

The choice of density separation fluid is equally important for the success of MP segregation from the matrix. Most researchers use sodium chloride (NaCl), which has a density of 1.2 g cm^{-3} as it is cost-effective; however, it has limited efficiency for high-density polymers. Also, it is not clear how many researchers have used analytical grade NaCl, since 94.4% of table salt is known to have MPs (Özçifçi et al. 2023). Some studies have used distilled water, sodium bromide (NaBr), calcium chloride ($CaCl_2$), zinc chloride ($ZnCl_2$), zinc bromide (ZnBr) and sodium iodide (NaI) to name a few. The efficacy of MP removal is quite high with NaI compared to NaCl, but its cost is prohibitive, approximately 70 times costlier than NaCl in some countries. While $ZnCl_2$ is not very environmentally friendly, it has been observed that some research groups do a density separation in a single step, e.g., a study from Mauritius using NaCl, while others have done a sequential separation, repeating the density separation several times using $ZnCl_2$ or NaI, e.g., the study from Chile and Ontario (Table 3.1).

Organic matter digestion can be achieved through various chemical methods, including acids, hydrogen peroxide (H_2O_2), and Fenton's reagent. The use of 65% nitric acid (HNO_3) or H_2SO_4 effectively removes OM, achieving <1% residual organics after 24 hours at 90 °C, but it risks degrading polymers like acrylonitrile butadiene styrene (ABS), polyamide (PA), and polyethylene terephthalate (PET). Among the studies reviewed, a Swiss floodplain study used acid digestion.

Various workers have also used alkaline solutions, such as a 1:1 KOH-NaCl mixture, which can mitigate plastic degradation while degrading OM.

In contrast, hydrogen peroxide (H_2O_2) is widely used for OM oxidation, with studies employing concentrations of 30–35% to disperse soil aggregates and digest organics, achieving up to 72% recovery for microplastic (MP) when applied to high-OM soils. For instance, 35% H_2O_2 effectively dispersed 72% of MPs bound to aggregates, though prolonged treatments (e.g., 72 hours at 23 °C) may be required. This is the most commonly used method; several studies reported from China, Chile, and Greece (Table 3.1) have used H_2O_2 with temperatures ranging from 50–70 °C and reaction time varying between 24 and 72 h. This reaction is exothermic, and it is highly recommended that samples be put in an ice bath to lower the temperature and preserve the polymer integrity.

Fenton's reagent, an advanced oxidation method combining H_2O_2 with Fe^{2+} under acidic conditions (pH 3–5), generates hydroxyl radicals that degrade OM with near-complete efficiency (100 ± 10%) at ambient temperatures. This method has shown promise in organic-rich sediments, recovering 90% of polyethylene (PE) without high-temperature inputs, though it remains under evaluation for broader soil MP separation applications. Fenton's reagent provides high degradation efficacy under mild conditions; each method requires context-specific optimization to minimize plastic damage and maximize organic clearance. The studies from Ontario, Canada, and Shouguang, China (Table 3.1) used Fenton's reagent for recalcitrant organics. Further research is needed to standardise these protocols, particularly for Fenton-based approaches.

The filtration of soil samples for microplastic analysis is a critical step that requires careful selection of filter types, pore sizes, and polymer compositions to ensure efficient capture and accurate identification. Membrane filters, commonly fabricated from polymers such as polycarbonate, nitrocellulose (or mixed cellulose esters), polytetrafluoroethylene (PTFE), and nylon, are widely employed due to their distinct chemical and physical properties. Polycarbonate membranes are favored for their smooth, uniform surface and high chemical resistance, facilitating clear microscopic imaging and Fourier-transform infrared (FTIR) or Raman spectroscopy without spectral interference. Nitrocellulose filters, while cost-effective and suitable for routine use, may require caution due to potential biodegradability in harsh chemical treatments. PTFE membranes, known for their hydrophobicity and exceptional resistance to aggressive reagents, are ideal for organic solvent-based digestion processes. Pore sizes typically range from 0.45 μm to 5 μm, with 0.45 μm being standard for capturing small microplastics (≥1 μm), though larger pores (e.g., 1.2–5 μm) may be used to prevent clogging from soil particulates, sometimes in tandem with pre-filtration steps. The choice of pore size balances the need to retain

target microplastics while managing sample load and filter longevity. The choice of pore size should also take into account the techniques to be used for identification, i.e., if an ATR-FTIR is to be used, MPs <300 μm are usually not attempted, while if micro-Raman spectroscopy is used, technically one can identify polymer up to 1 μm size.

Polymer selection also considers compatibility with downstream analytical techniques; for instance, nylon's hydrophilic nature suits aqueous solutions, whereas PTFE's inertness avoids spectral overlap with common synthetic polymers like polyethylene (PE) or polypropylene (PP). If Nile Red solution is to be used for staining the MPs, a polymer filter is to be avoided. Glass fiber filters, though non-polymeric, are occasionally used for high-throughput sieving but lack the precision of polymer membranes. Among the studies reviewed, the filter type is documented whenever provided in Table 3.1. In the study from Switzerland cellulose acetate filter with 0.45 μm pore size was used, which is prone to degradation during processing. The study indicated that nanoplastics were captured, but the risk of clogging was high. On the other hand, the studies from China have used nylon membrane filters and glass fiber filters in a wide range of pore sizes from 0.45 to 25 μm. High-quality nylon filters of 20 μm were shown to be efficient in a study from Shaanxi, China, whereas in another study from Xinjiang, China, the large-pore filters (>50 μm) were reported to have probably underestimated small MPs. Ultimately, the interplay of filter material, pore size, and soil matrix characteristics ensures optimal recovery of microplastics while minimising contamination and analytical interference, crucial for reliable environmental monitoring.

3.4 Microplastic Identification

The identification of MPs is usually done in two steps: a microscopic examination followed by a polymeric characterization using one of the chemical characterization approaches.

3.4.1 Microscopic Characterization

The microscopic identification of MPs is the most prevalent approach using a compound, dissection, stereomicroscope, or UV microscope. It provides a first overview of the likely MP's presence in a sample based on visual inspection and physical characteristics (Uddin et al. 2020a, 2022a). The absence of cellular structure, consistency in thickness along the length of the particle, consistent colour, no bending or tapering, clarity, and higher relief to name a few are characteristics of a polymer particle. Based on the microscopic examinations, the shape of possible MPs is also identified as fibre, fragment, film, and sphere. Several researchers have highlighted the use of fluorescent staining dyes such as Nile Red (Hidalgo-Ruz et al. 2012) to

improve the detection of MPs under a UV microscope. However, it has been shown by subsequent workers that the staining is not exclusive to MPs only; a lot of natural materials like wood, wool, chitin, and natural fibres also are stained, leading to significant misidentification (20–70%) of smaller MPs <100 μm (Song et al. 2015). In the reviewed studies (Table 3.1) most of them have used microscopic examination. Some have done morphotyping of MPs into fibres, fragments, and films, like the study from Shanghai, China. In another study from Murcia, Spain, the MP size determination was done using ImageJ software. Despite its accessibility and low cost, microscopic identification alone is considered insufficient for definitive MP identification and has resulted in false positives where organic fibers have frequently been misidentified as MPs.

3.4.2 Polymeric Characterization of Microplastics

To overcome the issue of false positives, it was realised that polymer validation is crucial (Shim et al. 2017). The most commonly applied techniques for polymer validation are Fourier-transform infrared (FTIR) and micro-Raman spectroscopy (Mariano et al. 2021). Additionally, automated digital imaging systems and machine learning algorithms are increasingly integrated to reduce human bias and enhance throughput (Bianco et al. 2020; Beres et al. 2023).

3.4.2.1 Fourier-Transform Infrared (FTIR) Spectroscopy

The Fourier-transform infrared spectroscopy identifies polymers by detecting bond-specific infrared absorption, differentiating MPs from organic or inorganic contaminants (Loder et al. 2017). The common operational modes in FTIR include reflectance and transmission, where the transmission mode requires thin, transparent samples, whereas the reflectance mode can work with irregularly shaped and opaque particles (Fok et al. 2020). In the study from the Netherlands, the transmission mode was used, and the standard particle size was >500 μm (Table 3.1). The most commonly used FTIR for microplastic analyses is attenuated total reflectance-FTIR (ATR-FTIR) which is quite effective for larger particle sizes (≥300 μm) and was used by the study conducted in Greece (Table 3.1). In the ATR-FTIR, we have to put the sample manually on the holder, and it is quite cumbersome if a large number of MPs are present in the sample. The micro-FTIR is coupled with microscopy, and it can effectively identify particles as small as 10 μm. Current models of micro-FTIR often allow semi-automated mapping and can streamline workflows and improve throughput. The data generated are often referenced against spectral libraries, and different workers use different thresholds; the most commonly accepted threshold is ≥70% match.

3.4.2.2 Raman Spectroscopy

The use of micro-Raman spectroscopy enables identifying MPs down to 1 μm with micro-Raman systems (Kappler et al. 2016). The underlying principle behind the instrument leverages the Raman effect—frequency shifts in scattered light—to identify polymers by analyzing vibrational modes and molecular structures (Smith and Dent 2019). It generates unique spectral fingerprints non-destructively in most cases, but there are reports of MP burning from prolonged laser exposure. A study reported in Table 3.1 from the Tibetan Plateau used micro-Raman and detected pigments and additives as well. The technique struggles with fluorescent additives or pigments, necessitating rigorous sample purification. However, the fluorescence interference is often mitigated by applying baseline correction algorithms and using advanced detectors. Despite its strengths—high specificity, spatial resolution, and non-destructive nature—Raman spectroscopy faces limitations such as weak signal intensity, lengthy data processing, and complex sample preparation. These micro-Raman systems are more costly compared to an ATR-FTIR and micro-FTIR.

3.4.2.3 Pyrolysis-Gas Chromatography Mass Spectrometry (Py-GC/MS)

Pyrolysis-gas chromatography-mass spectrometry (Py-GC/MS) has been used by some research groups. The Py-GC/MS thermally decomposes MPs under oxygen-limited conditions, identifying polymers via their degradation products (Hermabessiere et al. 2018). Double-shot Py-GC/MS enhances efficiency by performing sequential thermal desorption at varying temperatures. Combined with pressurized liquid extraction (PLE), it achieves low detection limits (14 μg g^{-1} for tyre particles) and effectively quantifies PE, PP, PMMA, and PS (Okoffo et al. 2020). Another novel technique recently developed is the pyrolysis-gas chromatography coupled to Orbitrap-mass spectrometry (Py-GC/OMS) (Torres-Agullo et al. 2024). The Py-GC/OMS has been optimized for several polymers including polymethyl methacrylate (PMMA), cyclopentanone (nylon-66), polypropylene (PP), nitrile butadiene rubber (NBR), polyvinyl chloride (PVC), polyethylene terephthalate (PET), acrylonitrile butadiene styrene (ABS), polyethylene (PE), polycarbonate (PC), and polystyrene (PS) (Torres-Agullo et al. 2024). The use of Py-GC/MS allows the generation of data as polymer mass in the matrix and is often incomparable with the concentration data reported in number per unit mass or volume. The other issue is the technique doesn't provide insight into the size of MP or nanoplastic, which is again quite crucial for toxicity assessment. However, we have not seen any instance of its usage for MP determination in soil.

3.4.2.4 Laser Direct Infrared Spectroscopy (LDIR)

The emergence of LDIR as a transformative tool for microplastic research in the past few years is not only based on its ease of use and faster processing time, but also on its technological superiority. The LDIR utilizes a high-resolution camera

that is capable of resolving particles up to 10 μm and, coupled with the quantum cascade laser (QCL), its response time is quick and almost instantaneous. It acquires the entire infra-red spectrum for an MP in just one second (Isaac Chandran and Veerasingam 2025). The technique provides an excessive advantage in terms of time and sample preparation compared to any other conventional technique (FTIR, Micro-Raman or Py-GC/MS), making it very valuable for matrixes where there is a large number of MPs. Within the studies reviewed only the study from Shandong, China, has used LDIR for 20–1000 μm MPs.

3.4.2.5 Thermal Analysis

Thermal analyses such as thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) are destructive techniques that are used for MP characterization (Isaac Chandran and Veerasingam 2025). These techniques chemically characterize MPs by assessing thermal degradation profiles (Dümichen et al. 2017). The efficacy of the thermal analysis is limited in differentiating between similar polymers, like PVC, polyurethane, PET, and polyesters. Despite these constraints, DSC remains advantageous for its simplicity and rapid results.

3.4.2.6 Scanning Electron Microscopy (SEM)

The scanning electron microscope (SEM) paired with the energy-dispersive X-ray spectroscopy (EDS) is an imaging technique that analyzes MPs' surface morphology and elemental composition (Sadri and Thompson 2014). However, its application is limited by high costs, time-consuming protocols, and challenges in identifying non-distinct particles (Primpke et al. 2020) making it less suitable for routine monitoring. In a study reported within this review, a study from Tamil Nadu, India, used SEM-EDS for studying the surface morphology and elemental composition of MPs. The complex sample preparation, including drying, and conductive coating the sample, can obliterate the MP morphology as well. It is a destructive technique with limited capability of polymer identification.

3.4.2.7 Liquid Chromatography-Mass Spectrometry (LCMS)

Liquid chromatography-mass spectrometry (LC-MS) has emerged as a powerful analytical tool for identifying and quantifying microplastics, sub-micron particles, or chemically degraded polymers. The technique leverages chromatographic separation to isolate analytes, followed by mass detection, often employing electrospray ionization (ESI) for polar polymers or additives like phthalates. However, LC-MS enables targeted analysis of polymer-specific degradation products, additives, or monomers, offering high sensitivity and specificity in complex environmental matrices. While LC-MS excels in detecting dissolved or chemically extracted

components, its application requires rigorous sample preparation, including enzymatic or oxidative digestion to remove organic interference without altering target polymers. It is prudent to highlight the challenges in universal polymer applicability and its excellent ability to detect PET and PS; LC-MS complements existing methods by providing quantitative insights into subvisible particles and additive leaching, advancing environmental monitoring and toxicological studies of microplastics. The information provided will not include size and shape data and will be comparable to information derived from Py-GCMS.

3.4.2.8 Atomic Force Microscopy (AFM)

The AFM generates nanometer-resolution images by scanning samples in contact or non-contact modes. When integrated with Raman or IR spectroscopy, it provides simultaneous topographic and chemical data. For instance, AFM-IR detects thermal expansion from IR absorption, translating it into frequency patterns via Fourier analysis that can be used for MP identification (Dazzi et al. 2017). However, it is not very commonly used until now.

3.5 Quantification of Microplastics and Reporting Units

The quantification of MPs in the absence of standardised methodologies and reporting units has introduced significant challenges in data comparability and interpretation. Researchers employ diverse analytical techniques and units of measurement, leading to the generation of incomparable datasets, making any spatio-temporal assessments challenging and complicated. This issue is particularly pronounced in soil and agricultural systems, where heterogeneous matrices amplify methodological disparities. Despite these issues, most researchers, about 80% of the studies reviewed have reported MP abundance in terms of MP Kg^{-1}. Among the studies reviewed in this paper, a study from Taiwan reported MP abundance in surface soil as MP m^{-2}. In some studies, like from Xinjiang, China, where mulch was added to soil, the MPs are reported as g m^{-2}. The results from studies where Py-GC/MS, LCMS, and TGA are used results are often reported as mg Kg^{-1}, though they are not frequently used for MP in soils.

3.6 Quality Assurance and Quality Control

Robust quality assurance and quality control (QA/QC) protocols are essential to ensure the integrity of microplastic (MP) analyses across all stages, from field sampling to laboratory processing and identification. Contamination risks—arising from airborne MPs, laboratory equipment, or reagents—must be systematically

minimized to safeguard the precision and validity of results. To address these challenges, stringent measures should be implemented, including the use of metal tools and avoiding plastic materials. Most of the studies reviewed in this paper have avoided using plastic tools. Several workers have reported thorough cleaning of instruments with Milli-Q water and used cotton lab coats. Most workers have covered the samples with aluminium foil to minimize airborne contamination during sample preparation, and few have used laminar flow cabinets and maintained clean-air environments, supported by high-efficiency particulate air (HEPA) filtration systems to reduce airborne contamination (Schell et al. 2022; Möller et al. 2020). Procedural blanks, including container and solution blanks, are processed identically to samples to monitor background contamination. About 90% of the studies reviewed in this communication have used blanks. Additionally, recovery tests using spiked samples have been used by many research groups. High recoveries of PE/PP (80–95%) have been reported by workers using $NaCl/ZnCl_2$, while low recovery was reported for PET/PVC, essentially as they used NaCl and these polymers are denser than the separating fluid; one such report was from Shanghai, China. Another important aspect of QA/QC is the replicate samples. The study from Germany and Spain analyzed triplicate samples, whereas Tanzania and Mexico analysed a single sample. The QA/QC strategies are to be strictly followed to enhance the reliability of MP inventories.

3.7 Methodological Challenges and Recommendations

There have been significant differences among studies in sample collection, preparation, and identification. These differences are evident since the soil is a heterogeneous matrix, and also because there is a huge disparity in the research resources and scientific questions addressed.

Based on the studies reviewed, it is observed that studies need to adopt standardized protocols; some of the key considerations to be taken are summarized below.

3.7.1 Sampling Strategies

Since soil is a heterogeneous matrix, collection of representative samples is a key to generating realistic data. It is highly recommended to undertake composite sampling by combining multiple subsamples across the study area. In agricultural systems, where there is high plastic input due to the use of mulch, sludge application, compost, or use of wastewater for irrigation, due consideration should be given not to skew the sampling to the topsoil only. In order to capture the true extent of MP pollution in agrosystems, it is recommended that stratified samples from 0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm be collected to assess vertical distribution, especially in tilled fields. Depending on the size of the study area, the sampling can

either be grid or random. It is extremely important to note down the GPS coordinates and all relevant information on agricultural practices like tillage, crops grown, irrigation practices, and whether soil additives, sludge, mulch, etc., have been used.

3.7.2 Extraction and Separation

From the published papers it is explicit that most studies have done a density separation using distilled water, NaCl, NaI, sodium polytungstate (NaPT), or zinc chloride ($ZnCl_2$). The recovery rates with distilled water are expected to be very low due to the negative buoyancy of most polymers. However, density separation is very high with $ZnCl_2$ but it is corrosive for high recovery rates. We recommend NaI though it is much more expensive than NaCl. It has also been observed that centrifugation enhances MP removal efficiency. Additionally, sonication has proven to be beneficial, but it should be low-energy sonification else MPs can also disintegrate into smaller sizes. For the digestion of organic matter from soil, 30% H_2O_2 is quite effective. However, KOH or enzymatic digestion with cellulase or protease is effective and would minimize polymer degradation. For clay-rich soils, adding a dispersant like sodium hexametaphosphate during extraction improves the extraction of MPs.

3.7.3 Identification and Quantification

Most MP research groups use a multi-threaded approach since the mere microscopic examination is not sufficient for the identification of tiny plastic particles. The first step toward the MP identification is microscopic examination using a stereo-microscope, compound microscope, dissection microscope, or UV microscope to identify the morphology (size, shape, colour). However, for polymer identification, one of the several analytical techniques discussed in this paper can be used; the most commonly used polymer identification techniques are ATR-FTIR (>300 μm), micro-FTIR (20 μm), and micro-Raman spectroscopy (1 μm).

Some advanced techniques like Py-GC/MS, Py-GC/OMS, and LC-MS can provide mass quantification of MPs and even for nanoplastics. However, they have limited usage in MP quantification in soil; it might have high usage if food chain transfer from soil to edible plants is to be looked at.

3.7.4 Data Reporting

The data is usually reported as particle count per unit mass of matrix, which is the soil in this case; thus, the results are to be reported as MP Kg^{-1}. Some research groups also report mass concentrations as mg Kg^{-1}, but that is only feasible in soils

where high MP concentration is seen (i.e., sludge applied or mulched soils). It is important to report the size range of MPs as well.

In the case of soil, the metadata is also important, as details of soil texture, grain size, organic matter content, agricultural inputs (fertilizers, irrigation), and coordinates should be recorded.

3.7.5 Quality Control

Looking at the wide range of methodologies and practices employed for MP determination in soils, the QA–QC becomes more so important to generate a reliable dataset. Use of cotton lab coats, glass and metal containers, and working in a laminar flow unit is highly recommended. Also, the use of both field and lab blanks will provide information for realistic correction of data. Some of the studies have used spiked samples to validate methods, and it is also highly recommended.

3.8 Conclusion

This summary of studies so far on microplastics in soil highlights the limited work carried out so far, even though gigantic amounts of MPs end up in agricultural lands as a result of the use of sewage, compost for amending low-productivity soils, use of wastewater for irrigation, and mulches for improved productivity. Only 39 studies were published where details of the methodology are reported for MPs in soil. A quick overview highlights that the methods used for soil are modified methods for MP determination in marine and freshwater matrixes. One very conspicuous difference between soil and other matrixes is the heterogeneity of soil, which makes it imperative that representative samples are collected. The presence of fine-grained clastics, mainly clays, warrants specific steps to ensure the segregation of MPs from the matrix. The digestion of organic matter using 30% H_2O_2 or enzymatic digestion seems quite effective. The use of Py-GC/MS and LCMS for MPs in soil gives a very different level of information on the mass of polymers rather than number. Though not very popular, we have found LDIR to be very effective in the detection of MPs; it provides both the size information and numbers and type of polymers with reasonably high accuracy. There is certainly a need to harmonize methods to be used for MPs in soil for generating comparable datasets across spatial domains. Future work should address nanoplastics and biodegradable polymers, ensuring methodologies evolve with agricultural plastic use.

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